

Quiz 1

Started: Jan 18 at 9:19am

Quiz Instructions

Intro and Universal Approximators

This quiz covers lectures 1 and 2. Several of the questions invoke concepts from the hidden slides in the slide deck, which were not covered in class. So please go over the slides before answering the questions.

You will have three attempts for the quiz. Questions will be shuffled and you will not be informed of the correct answers until after the deadline. While you may discuss the concepts underlying the questions with others, you must solve all questions on your own - see course policy.



Question 1 1 pts

According to the university's academic integrity policies and 11785 course policies, which of the following practices are **NOT** allowed in this course? Select all that apply.



Discuss quiz solutions with another student before the deadline



Creating a piazza post asking for help debugging code without prior attempts to debugging it yourself.



Asking a TA for help debugging your code



Discussing concepts from class with another student



Posting code in a public post on piazza



Helping another student debug their code



Question 2 1 pts

Is the following statement true or false? Hebbian learning allows reduction in weights and learning is bounded.

Slide: lec 1, "Hebbian Learning" slide 66

☐

True

☒

False



Question 3 1 pts

Which of the following are **NOT** part of Aristotle's laws of association? (select all that apply)

Hint: See lec 1: Slides on "Associationism" 32-37

☐

Thoughts about one event or thing triggers thoughts about related events or things.

☒

Thoughts about one event or thing triggers thoughts about events or things that have opposite qualities

☒

Mental willpower is used to associate unrelated events or things

☒

Associations are formed from logical deduction.

☐

Events or things near the same place and time are associated.



Question 4 1 pts

How does the number of weights in an XOR network with arbitrarily many hidden layers and threshold logic gates grow with the number of inputs to the network?

See lec 2: Slides on "Optimal depth" and "Network size" 113-123

☐

Polynomial but faster than linear

☐

Linear

☒

Between polynomial and exponential

☐

Exponential or faster



Question 5 1 pts

What is the implication of Shannon's theorem on network size (as a function of input size) for Boolean functions?

Slide: lec 2, slide 71-72

- ☐ Only a relatively small proportion of functions require exponentially sized inputs
- ☐ Most functions require polynomial sized networks.
- ☐ All functions require exponential-sized networks
- ☒ Nearly all functions require exponential-sized networks



Question 6 1 pts

What is the fewest number of neurons needed (including any output layer neurons) for a network to implement the truth table shown by the following Karnaugh map? (numeric answer, int and float are both fine)

WX \ YZ		YZ			
		00	01	11	10
WX	00				
	01				
	11				
	10				

Hint: lec 2, "Reducing a Boolean Function". Slide 47 - 50



Question 7 1 pts

Suppose the data used in a classification task is 10-dimensional and positive in all dimensions. You have two neural networks. The first uses threshold activation functions for hidden layers, and the second uses softplus activation functions for hidden layers. In both networks, there are 2000 neurons in the first hidden layer, 8 neurons in the second hidden layer, and a huge number of neurons for all later layers. It turns out that the first network can never achieve perfect classification. What about the second network?

Hint: A layer with 8 neurons effectively projects the data onto a 8-dimensional surface of the space. The input is 10-dimensional.

Lec 2 slides 138-150



Assuming that layers 3 and above are so expressive that they never bottleneck the flow of information, the second network will be able to achieve perfect classification.



It might fail for some data sets, not only because a bottleneck can occur, but also because the 2000 neurons in the first hidden layer could bottleneck the flow of information if the classification task is sufficiently complex.



It too is guaranteed to be unable to achieve perfect classification.

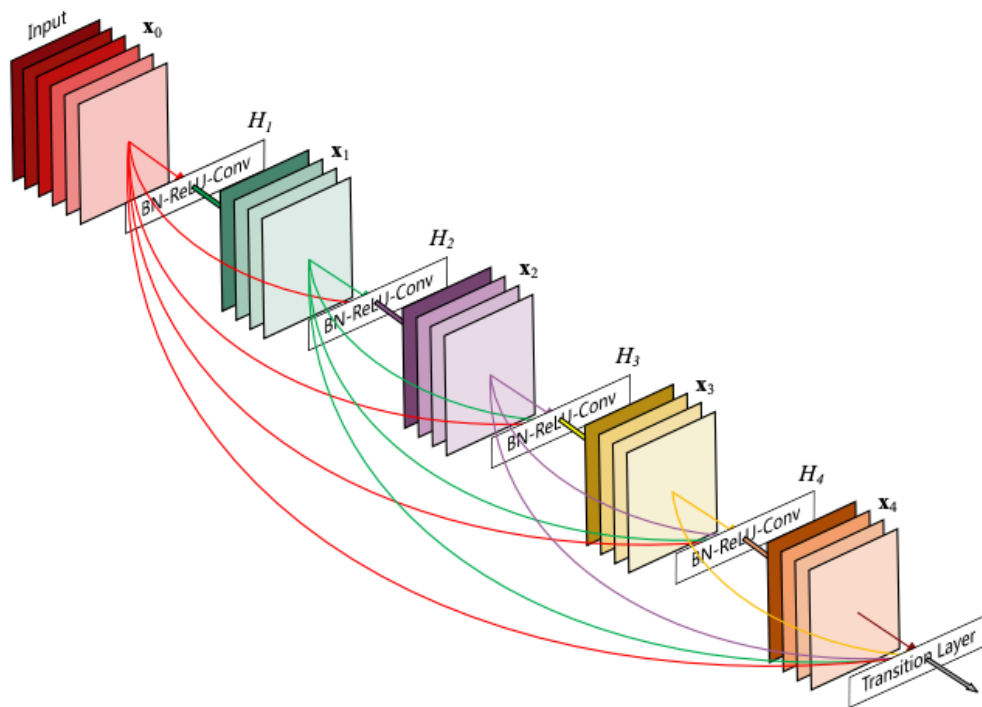


It might fail for some data sets, since the 8 neurons in the second hidden layer could bottleneck the flow of information. In that case, the sizes of layers 3 and above don't matter.



Question 8 1 pts

Densenet (2017) is a CNN network architecture that achieved SoTA results with fewer params than many of its contemporaries. Its main idea was the 'dense block': a block where each layer output is concatenated into the input of the downstream layers.



Above is a diagram of a dense-block. Each " H_i " consists of Batchnorm->ReLU->Convolution. Each H_i outputs a " x_i " (one stack of squares in the image) .

What is the depth of the dense block above? (numeric answer, int and float are both fine)

Assume that "BN+RELU+CONV" is a single layer. The square planes simply represent the data flowing down between layers and are not layers themselves.

(Note: for the definition of network depth, see the lecture 2 recording)

Slide: lec 2, "Deep Structures"

4

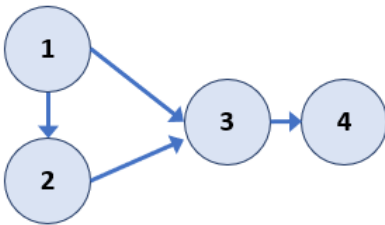
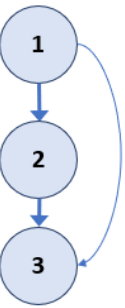
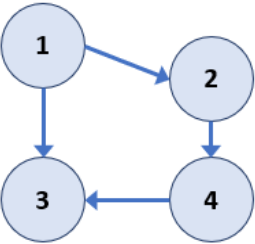
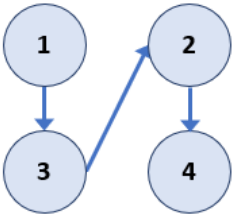


Question 9 1 pts

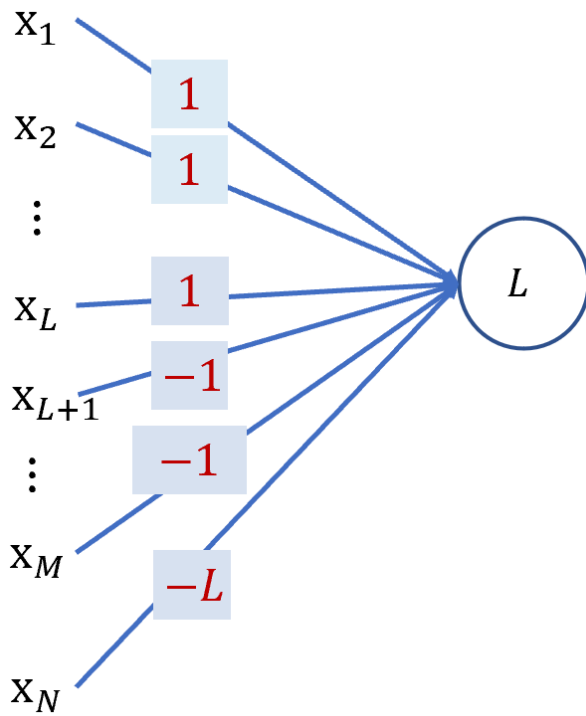
Which of the following NNs have a depth of 3? (Choose all that apply)

(Note: for the definition of network depth, see the lecture 2 recording)

Hint: lec 2, "Deep Structures", Slides: 17-18



Question 10 1 pts



What boolean function does the above perceptron represent?

(Hint: notice that $M = N - 1$)

Hint: Lecture 2 Slides: "Perceptron as a Boolean gate", Slides 26-30

☐

$$\left(\bigvee_{i=1}^L X_i \right) \wedge \left(\bigwedge_{j=L+1}^M \bar{X}_j \right) \vee (X_N)$$

☐

$$\left(\bigwedge_{i=1}^L X_i \right) \wedge \left(\bigwedge_{j=L+1}^M \bar{X}_j \right) \vee (X_N)$$

☐

$$\left(\bigwedge_{i=1}^L X_i \right) \vee (X_M) \wedge (\bar{X}_N)$$

☒

$$\left(\bigwedge_{i=1}^L X_i \right) \wedge \left(\bigwedge_{j=L+1}^N \bar{X}_j \right)$$

Quiz saved at 2:15pm

Submit Quiz